

Pratik Kumar Padhi

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SUMMARY

AI/ML engineering student specializing in computer vision and agricultural AI, with hands-on experience building and deploying end-to-end deep learning pipelines. Conference presenter at ICCAIML 2026 with co-authored research on AI-driven crop health monitoring. Proficient in Python, PyTorch, TensorFlow, and AWS deployment. Seeking internship roles in AI/ML, Data Science, or Software Engineering to contribute research depth and production-level project experience.

EDUCATION

Centurion University of Technology and Management Bhubaneswar, Odisha
B.Tech in Computer Science and Engineering (AI & Machine Learning Specialization) Aug 2023 – May 2027

- Relevant Coursework: Machine Learning, Deep Learning, Data Structures & Algorithms, Database Management Systems, Computer Vision, Statistics for Data Science

TECHNICAL SKILLS

Languages: Python, SQL, C++, Java, JavaScript
ML/AI Frameworks: PyTorch, TensorFlow, Keras, scikit-learn, OpenCV, YOLOv8, Hugging Face
Data & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Power BI, DAX, Jupyter Notebook
Cloud & DevOps: AWS (EC2, S3, Lambda), Git, GitHub, Linux, Docker (basic)
Web: HTML, CSS, React (basic), Flask/FastAPI, Netlify
Enterprise: SAP S/4HANA, ABAP, SAP Fiori, Power Query

PROJECTS

Cropsensy — AI-Powered Crop Prediction Platform | *Python, TensorFlow, AWS, Flask* | [Live Demo](#) 2024 – Present

- Engineered an end-to-end crop yield prediction pipeline using a CNN-LSTM hybrid model trained on multi-variable soil, weather, and environmental datasets, achieving 92% classification accuracy on held-out test data.
- Deployed the full-stack application on AWS EC2 + S3 with a Flask REST API backend, achieving 99.5% uptime over a 6-month production window.
- Implemented automated data preprocessing and feature engineering pipeline reducing model retraining time by 30%, enabling weekly model updates with fresh sensor data.

Real-Time Weed Detection System | *YOLOv8, PyTorch, OpenCV, Python* Jan 2025 – Apr 2025

- Built a real-time agricultural weed classification system using YOLOv8 fine-tuned on a custom-annotated dataset of 12 weed species, achieving 94% mAP@0.5 on the validation set.
- Optimized inference pipeline with TorchScript model export and batch processing, reducing per-frame latency from 120ms to 72ms (40% improvement), enabling real-time field deployment on edge hardware.
- Presented findings at ICCAIML 2026 and co-authored a research paper demonstrating 25% improvement in detection reliability over traditional threshold-based methods.

Coral Reef Health Classification System | *TensorFlow, Keras, OpenCV, Python* Aug 2024 – Dec 2024

- Developed a 3-class image classification model (healthy / bleached / degraded) using a ResNet-50 transfer learning architecture fine-tuned on 8,000 labeled underwater images, achieving 94% test accuracy.
- Built an automated monitoring dashboard with real-time image ingestion and anomaly alerting, reducing manual coral assessment review time by 70% in pilot testing.
- Optimized image preprocessing pipeline using OpenCV and NumPy vectorization, achieving 12× throughput increase over the baseline sequential processing approach.

Uber NCR Analytics Dashboard | *Power BI, DAX, Power Query, Data Modeling, Bing Maps* 2025

- Built an end-to-end Power BI dashboard analyzing 150K+ Uber ride bookings across Delhi NCR, tracking 51.85M in revenue, 62% ride completion rate, and 4.59M in lost revenue across 176 pickup locations.
- Modeled multi-table data relationships using Power Query and DAX measures to surface vehicle-wise performance, driver/customer rating trends, and peak-hour demand patterns.
- Implemented geospatial intelligence layer using Bing Maps integration, identifying high-value pickup clusters and uncovering that a small number of locations drove a disproportionately high share of total bookings.

ICCAIML 2026 — International Conference on AI and Machine Learning

Presenter & Co-Author: “Deep Learning for Agricultural Automation”

2026

- Co-authored peer-reviewed research demonstrating 25% improvement in crop health monitoring accuracy using deep learning over traditional image processing baselines.

SAP Certified Associate — Business Process Integration with SAP S/4HANA Issued by SAP

Credly Verified Badge

2025

- Certified in end-to-end business process integration across SAP S/4HANA modules; covers core ERP workflows and cross-functional process design.

SAP Certified — Back-End Developer (ABAP Cloud) Issued by SAP

Credly Verified Badge

2025

- Certified in ABAP programming, RESTful Application Programming Model (RAP), and Clean Core principles for building custom SAP S/4HANA extensions.

SAP Certified — SAP Fiori Application Developer Issued by SAP

Credly Verified Badge

2025

- Certified in SAP Fiori frontend development; covers UI5 application architecture, Fiori design principles, and SAP BTP integration for enterprise app delivery.

Also: Java Programming — GeeksforGeeks (2026) | Target: AWS Cloud Practitioner (in progress)